# AIR POLLUTION AND CONTROL

## Subject Code : 23A01504b

## UNIT - III : CONTROL OF PARTICULATE POLLUTANTS

**Syllabus:** Properties of particulate pollution - Particle size distribution - Control mechanism - Dust removal equipment - Design and operation of settling chambers, cyclones, wet dust scrubbers, fabric filters and ESP.

## I. Properties of Particulate Pollutants

The choice and design of control equipment depends on the **properties of the particles** and of the carrier gas.

### I.I Particle properties

- **Size (diameter):** most important property; ranges from 0.001 um to > 100 um. For irregular particles an equivalent diameter is used: aerodynamic, Stokes, projected-area diameter
- **Aerodynamic diameter:** diameter of a unit-density (1 $g/cm^3$) sphere having the same settling velocity as the particle
- **Shape:** spherical, fibrous, flakes, irregular; non-spherical particles settle slower (shape factor)
- **Density:** true (particle) density and bulk density; settling velocity increases with density
- **Specific surface:** surface area per unit mass - high for fine particles; adsorbs gases, high reactivity
- **Cohesion / adhesion:** sticky or hygroscopic dust blocks filters and sticks to surfaces; very fine dust agglomerates
- **Electrical properties:** charge acquired and **electrical resistivity** (important for ESP)

- **Optical properties:** scatter light (visibility, opacity) - maximum scattering for 0.1 - 1 um
- **Hygroscopicity, flammability, explosivity, toxicity, abrasiveness, corrosiveness** - influence the material of construction and safety

### 1.2 Gas properties

- Temperature, pressure, humidity, **dew point**, viscosity, density, flow rate, composition (acid gases), dust loading ($g/m^3$)

| Size range (um) | Behaviour in air | Typical particles |
|---|---|---|
| > 10 | Settle fast by gravity; removed in nose | Dust, sand, fly ash coarse |
| 1 - 10 | Settle slowly; impaction | Fly ash, cement, pollen |
| 0.1 - 1 | Stay airborne for days; best light scattering | Smoke, fumes, acid mist |
| < 0.1 | Brownian motion, diffusion; coagulate | Fumes, combustion nuclei |

## 2. Particle Size Distribution

A dust sample contains particles of many sizes. Its **particle size distribution** shows the fraction of mass (or number) in each size range. It is measured by sieving, microscopy, sedimentation, cascade impactor, laser diffraction, coulter counter etc.

- **Frequency distribution:** % mass in each size interval (histogram)
- **Cumulative distribution:** % mass **smaller than** (undersize) or **larger than** (oversize) a given size; plotted on **log-probability paper**
- Most air-borne dust follows a **log-normal distribution** - a straight line on log-probability graph

### Important parameters

- **Mass median diameter (MMD or $d_{50}$):** diameter above and below which 50% of the mass lies (read at 50% on the cumulative plot)
- **Geometric mean diameter:** $d_g = (d_1^{n1} \times d_2^{n2} \times \dots)^{1/N}$ or $\log d_g = \text{sum}(n_i \log d_i) / N$
- **Geometric standard deviation:** $\text{sigma}_g = d_{84.1} / d_{50} = d_{50} / d_{15.9}$ ($\text{sigma}_g = 1$ for monodisperse, > 1.2 polydisperse)
- **Number average diameter** $d_n = \text{sum}(n_i d_i) / \text{sum}(n_i)$; **surface mean** and **volume mean** diameters
- **Hatch-Choate relations** connect the different mean diameters of a log-normal distribution: $\text{MMD} = \text{CMD} \times \exp(3 \ln^2 \text{sigma}_g)$ (CMD = count median diameter)

**Why needed? Collection efficiency of every device is a function of particle size - we need a fractional efficiency curve and the size distribution to find the overall efficiency: $\text{eta}_o = \text{sum}(\text{eta}_i \times m_i)$ where $m_i$ = mass fraction of size i.**

**Example:** three fractions with mass 20%, 50%, 30% and collection efficiencies of 40%, 80%, 95% -> $\text{eta}_o = 0.2 \times 40 + 0.5 \times 80 + 0.3 \times 95 = 8 + 40 + 28.5 =$ **76.5%**.

## 3. Control Mechanisms (Collection Mechanisms)

Particles are separated from the gas stream by applying a force that moves them to a collecting surface.

| Mechanism | Principle | Used in / best for |
|---|---|---|
| Gravitational settling | Gravity force on particle; $v_s$ | Settling chamber; > 50 um |
| Centrifugal force | Spinning gas throws particles to wall | Cyclone; > 5-10 um |

| Mechanism | Principle | Used in / best for |
|---|---|---|
| Inertial impaction | Particle cannot follow gas streamline around obstacle (droplet, fibre) | Scrubbers, filters; > 1 um |
| Interception | Particle following streamline touches the collector | Filters, scrubbers; 1 um |
| Diffusion (Brownian) | Random motion brings very small particles to the surface | Filters, scrubbers; < 0.3 um |
| Electrostatic attraction | Charged particle moves in electric field to electrode | ESP; 0.1 - 10 um |
| Thermophoresis, sonic agglomeration, condensation | Auxiliary mechanisms | Special cases |

## Stokes' law and settling velocity

**$v_s = g\, d^2\, (rho_p - rho_g) / (18\, mu)$ [valid for Re < 1, d in m]**

- $v_s$ = terminal settling velocity (m/s), d = particle diameter (m), $rho_p$ = particle density, $rho_g$ = gas density, mu = gas viscosity (air $1.8 \times 10^{-5}$ kg/m.s)
- $v_s$ proportional to $d^2$: a 10 um particle settles 100 times faster than a 1 um one
- For very small particles (< 1 um) use **Cunningham slip correction factor** $C_c > 1$: $v_s = C_c\, g\, d^2\, rho_p / 18\, mu$
- For Re 1 to 1000 (intermediate) use $v_s = [4\, g\, d\, (rho_p - rho_g) / (3\, C_D\, rho_g)]^{1/2}$ with $C_D = 24/Re \times (1 + 0.15\, Re^{0.687})$

**Example:** d = 50 um, $rho_p$ = 2000 kg/m$^3$: $v_s = 9.81 \times (50 \times 10^{-6})^2 \times 2000 / (18 \times 1.8 \times 10^{-5}) = 4.9 \times 10^{-5} / 3.24 \times 10^{-4}$ = **0.15 m/s**.

## 4. Dust Removal Equipment - Overview

| Equipment | Mechanism | Size removed | Efficiency | Pressure drop |
|---|---|---|---|---|
| Settling chamber | Gravity | > 50 um | 40 - 60% | 5 - 25 Pa |
| Cyclone | Centrifugal | > 5 um | 70 - 90% | 0.5 - 1.5 kPa |
| Wet scrubber (venturi) | Impaction, diffusion | > 0.5 um | 90 - 99% | 2 - 10 kPa |
| Fabric filter | Sieving, impaction, diffusion | > 0.1 um | 99 - 99.9% | 1 - 2 kPa |
| ESP | Electrostatic | > 0.1 um | 95 - 99.9% | 0.1 - 0.5 kPa |

**Selection depends on:** dust size and concentration, required efficiency, gas flow, temperature, corrosiveness, cost (capital + operating), space, availability of water, disposal of collected dust. Equipment are often used in series: **cyclone (pre-cleaner) + ESP / fabric filter (final cleaner)**.

## 5. Gravity Settling Chamber

A large rectangular chamber in which the gas velocity is reduced (1 - 3 m/s -> 0.3 m/s) so that the particles have enough time to settle. Used as **pre-cleaner** for coarse dust.

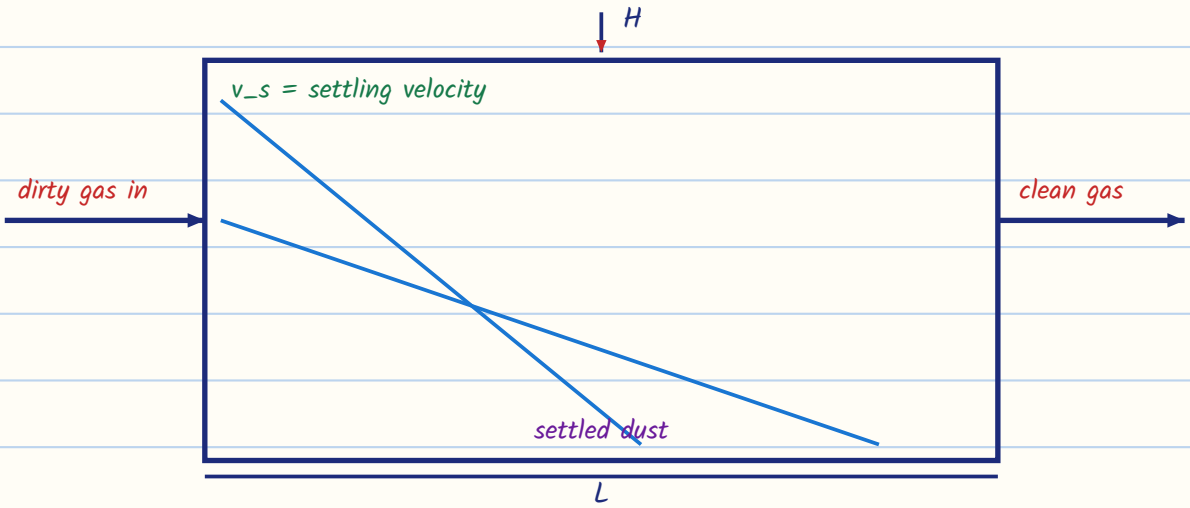



Fig. Horizontal-flow gravity settling chamber

## Design theory (laminar flow model)

- Residence time of gas: $t = L / v_h$ ($v_h$ = horizontal velocity = $Q / (W \times H)$)
- A particle at the top must fall height H in that time: $H / v_s <= L / v_h$
- **Efficiency (laminar / ideal, one size): $\eta = v_s L / (v_h H) = v_s L W / Q$** (100% when $v_s L / v_h H >= 1$)
- **Turbulent mixing model: $\eta = 1 - \exp( - v_s L W / Q )$**
- **Minimum size completely removed (cut size): $d_{min} = [ 18 \mu Q / (g \rho_p W L) ]^{1/2}$**
- Efficiency depends on the **plan area (L x W)**, not on the height H: use many trays (**Howard multiple-tray settling chamber**) - n trays give n times the efficiency (but cleaning is difficult)
- Design limits: $v_h$ < 3 m/s (to avoid re-entrainment of settled dust); typical $v_h$ 0.3 - 0.6 m/s; Re < 2000 for laminar flow; length is generally 2-3 times width
- Pressure drop very small; only for particles > 50 um (40-60% overall efficiency)

| Advantages | Disadvantages |
|---|---|
| Simple, cheap, no moving parts | Huge size / space needed |
| Low pressure drop, easy to maintain | Only coarse particles; low efficiency |
| Can handle high temperature and corrosive gases | Difficult to clean; re-entrainment |

**Example:** $Q = 1.5\ m^3/s$, $W = 3$ m, $L = 6$ m, $d = 50$ um ($v_s = 0.15$ m/s): $\eta = 0.15 \times 6 \times 3 / 1.5 = 1.8 > 1$, so **100%**; for $d = 20$ um ($v_s = 0.024$ m/s): $\eta = 0.024 \times 18 / 1.5 =$ **29%**.

# 6. Cyclone Separators

**Centrifugal force** (many times gravity) is used. Gas enters **tangentially** at the top of a cylindrical-conical vessel, spirals down as the **outer vortex**; particles are thrown to the wall, slide down into the hopper; the clean gas reverses and leaves as the **inner vortex** through the central outlet (vortex finder).

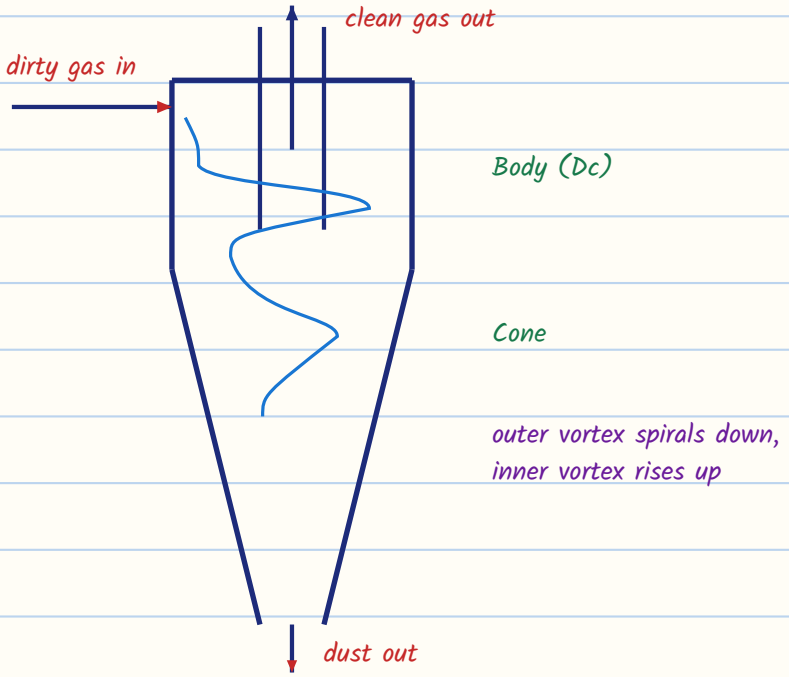



Fig. Reverse-flow (tangential inlet) cyclone

## Standard (Lapple) cyclone dimensions

| Symbol | Part | Dimension |
|---|---|---|
| $D_c$ | Body diameter | $D_c$ |
| $H_c$ | Inlet height (a) | $D_c / 2$ |
| $B_c$ | Inlet width (b) | $D_c / 4$ |
| $D_e$ | Gas exit (vortex finder) diameter | $D_c / 2$ |
| $S_c$ | Length of vortex finder | $D_c / 8$ |
| $L_b$ | Cylinder length | $2\ D_c$ |
| $L_c$ | Cone length | $2\ D_c$ |
| $J_c$ | Dust outlet diameter | $D_c / 4$ |

## Design equations

- Inlet velocity $V_i = Q / (H_c \times B_c)$; good practice **15 - 25 m/s** (below 15 poor efficiency; above 25 re-entrainment/erosion)
- **Number of effective turns:** $N = (1 / H_c)(L_b + L_c / 2)$ = about 5 for Lapple standard cyclone
- **Cut diameter (50% efficiency):** $d_{50} = [9 \, mu \, B_c / (2 \, pi \, N \, V_i \, (rho_p - rho_g))]^{1/2}$
- **Lapple fractional efficiency:** $eta_i = 1 / [1 + (d_{50} / d_i)^2]$
- **Pressure drop:** $dP = (rho_g \, V_i^2 / 2) \times K_c$, $K_c = 16 \, H_c \, B_c / D_e^2$ (about 8 velocity heads for standard design)
- **Overall efficiency** = sum of (mass fraction x fractional efficiency)

**Example:** $D_c = 0.5$ m, $Q = 0.5$ m$^3$/s -> $H_c = 0.25$, $B_c = 0.125$, $V_i = 0.5 / 0.03125 = 16$ m/s. With $N = 5$, $mu = 1.8 \times 10^{-5}$, $rho_p = 2000$: $d_{50} = sqrt(9 \times 1.8e\text{-}5 \times 0.125 / (2 \, pi \times 5 \times 16 \times 2000)) =$ **4.5 um**; $dP = 0.5 \times 1.2 \times 16^2 \times 8 =$ **1230 Pa**.

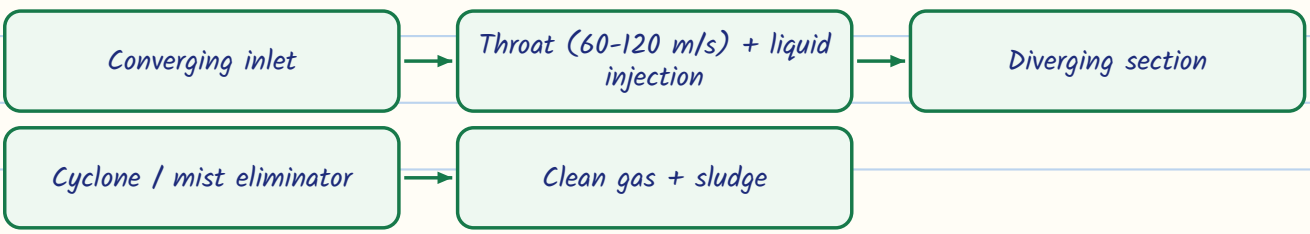
- **High-efficiency cyclones** (Stairmand) are tall and narrow with smaller inlet; **high-throughput** types have larger inlet; **multicyclones** (many small tubes in parallel) give 90% for 5 um
- Variants: cyclones with axial (straight-through) entry, reverse-flow, dust-return; **dry** and **wet (irrigated)**
- **Advantages:** cheap, no moving parts, compact, handles high temperature and pressure and heavy dust loads (up to 1000 g/m$^3$), low maintenance
- **Disadvantages:** poor for particles < 5 um, efficiency falls at low load, erosion in abrasive dust, high pressure drop and cannot handle sticky material

# 7. Wet Dust Scrubbers (Wet Collectors)

The dusty gas is brought into **intimate contact with liquid (water) droplets** or films; particles are captured by droplets (**inertial impaction, interception, diffusion**), then the droplets are removed by a mist eliminator. Capture of particles by impaction is given by the **target efficiency**, which rises with particle size, velocity and with small droplet size.

| Type | Working | Remarks |
|---|---|---|
| Spray tower | Water sprayed down, gas up; droplets 500-1000 um | Low dP (0.1-0.5 kPa); > 10 um only |
| Cyclone spray scrubber | Cyclone with water spray / wall film | Higher efficiency, > 2 um |
| Impingement plate / tray | Gas passes through holes onto plates, water on tray; froth | > 1 um; 1-2 kPa |
| Packed bed (tower) | Random packing wetted with liquid, counter/cross flow | For gas absorption + clean dust; clogging problem |
| Venturi scrubber | Gas accelerated in throat 60-120 m/s, liquid injected; atomised | Best for submicron; 2.5-15 kPa |
| Mechanical (rotor) scrubber | Power-driven rotor gives fine spray | High cost, energy use |

## Venturi scrubber




- Liquid : gas ratio **0.7 - 2 L / $m^3$** of gas; pressure drop 2.5 - 15 kPa, efficiency > 99% for 0.5 um dust at high dP

- **Johnstone equation: eta = 1 - exp( - k R sqrt(psi) )**; R = liquid / gas ratio ($m^3$ / $m^3$), k = correlation constant (0.1 - 0.2), **inertial impaction parameter psi = C rho$_p$ v$_t$ d$_p^2$ / (18 mu d$_0$)** where v$_t$ = throat velocity, d$_0$ = droplet diameter
- **Pressure drop** (Calvert) increases with throat velocity squared and L/G; **power consumption** is the main cost - 'collection efficiency vs energy' relation

| Advantages | Disadvantages |
|---|---|
| Collects both gas and particulate; cools hot gases | Water pollution: sludge disposal / treatment |
| High efficiency for fine and sticky dust | High energy use (venturi); corrosion and freezing |
| No explosion risk; compact; low cost | Visible steam plume; costly maintenance |

## 8. Fabric Filters (Bag House)

The dusty gas is passed through **woven or felted fabric bags** that retain the particles. Initially the cloth itself is the filter, but very soon a **dust cake** (layer) forms on the fabric surface which acts as the main filter medium - giving high efficiency even for submicron particles.

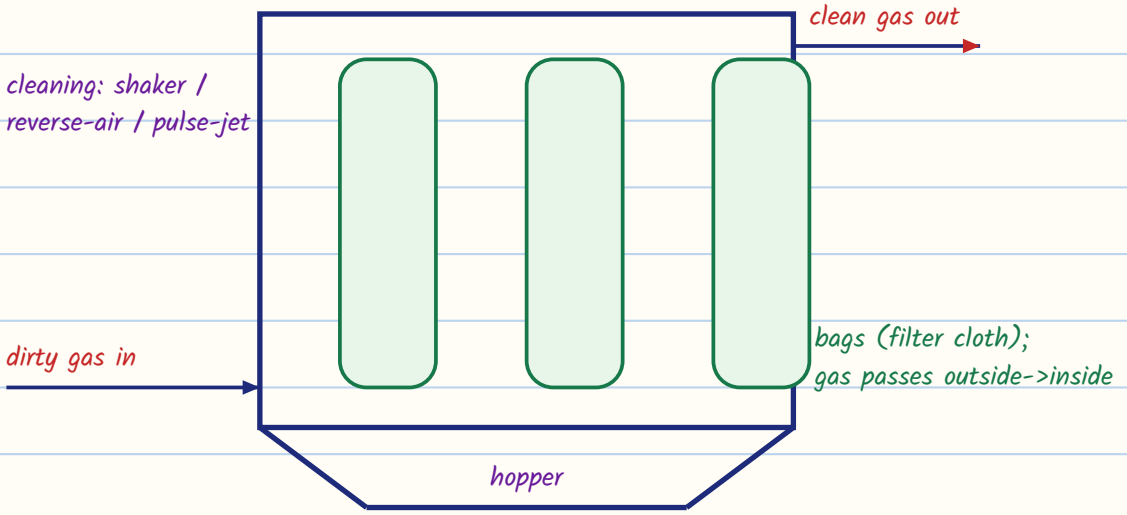


Fig. Baghouse (fabric filter) with cleaning arrangement

## Collection mechanisms

- **Sieving** by the cake (dominant), **inertial impaction** (> 1 um), **interception**, **diffusion** (< 0.3 um), **electrostatic** (charge on fibres and particles)

## Design

- **Air-to-cloth ratio (filtration velocity): $V_f = Q / A$** ($m^3$/min per $m^2$ = m/min). Typical 0.5 - 2.5 m/min (shaker 0.5-1, reverse air 0.5-1, pulse-jet 2-4)
- **Cloth area required: $A = Q / V_f$**; number of bags $n = A / (\pi D L)$ with bag diameter D = 0.1 - 0.3 m, length 3 - 10 m; add 10-20% for compartment being cleaned off-line
- **Pressure drop: $dP = dP_f + dP_c = K_1 V_f + K_2 C V_f^2 t$** ($K_1$ fabric resistance, $K_2$ specific cake resistance, C = inlet dust conc., t = filtering time) - typical 0.5 - 2 kPa (50 - 200 mm w.g.)
- Gas must be cooled above its dew point: temperature limit decided by the cloth

| Fabric | Max. temp (deg C) | Remarks |
|---|---|---|
| Cotton | 80 | Cheap; poor acid resistance |
| Wool | 100 | Good for acid; poor in alkali |
| Nylon / polyamide | 200 | Abrasion resistant; poor to acid |
| Polyester (Dacron) | 135 | Good general-purpose, poor in hot water |
| Nomex (aramid) | 220 | Good dimensional stability |
| Glass fibre | 260 | High temperature; weak to flexing |
| Teflon (PTFE) | 260 | Excellent chemical resistance; costly |

### Cleaning methods

1. **Mechanical shaking:** bag top oscillated; the filtering for that compartment is stopped. Simple; 'woven' fabrics
2. **Reverse air:** reverse gentle flow collapses the bag and cake falls; for glass fibre
3. **Pulse-jet:** short burst (0.1 s) of compressed air (5-7 bar) down the bag; **on-line** cleaning; felt; high ratio, compact

- **Advantages:** efficiency 99-99.9% for all sizes, handles variable loads, dry collection of product, simple and modular
- **Disadvantages:** limited temperature, moisture / sticky dust blinds the bag, fire hazard, bag life 1-3 years, large space for low-ratio types, high maintenance

## 9. Electrostatic Precipitator (ESP)

Uses **electrostatic forces** to separate charged particles. A very high DC voltage (40 - 100 kV) is applied between discharge electrodes (wires, negative) and grounded collecting plates, producing a **corona discharge** around the wire.

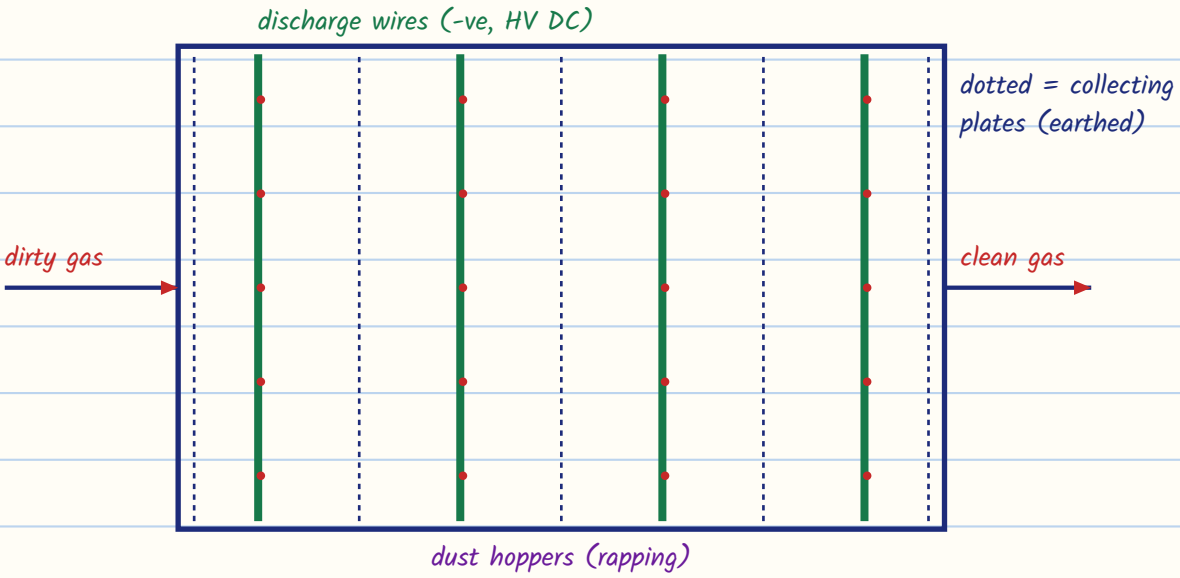



Fig. Plate-type electrostatic precipitator

## Working steps

1. **Ionisation:** corona produces gas ions (electrons and negative ions)
2. **Charging:** particles capture ions by field charging (> 1 um) and diffusion charging (< 0.2 um)
3. **Migration:** charged particles move towards the collecting electrode under electric field, drift velocity (migration velocity) w
4. **Deposition:** particles stick to the plates as a layer
5. **Removal:** plates are **rapped** (hammers / vibrators), dust falls to the hopper; for wet ESP the plates are flushed with water

## Design - Deutsch-Anderson equation

**eta = 1 - exp( - w A / Q ) = 1 - exp( - w L / (v d) ) -> A = - (Q / w) ln (1 - eta)**

- eta = collection efficiency, w = migration (drift) velocity (about 0.05 - 0.2 m/s), A = total collecting plate area ($m^2$), Q = gas flow ($m^3/s$)
- **Migration velocity: w = q $E_p$ / (3 pi mu d)** (q = particle charge, $E_p$ = field strength at collecting plate) - w proportional to field strength squared and particle size
- **Specific collecting area (SCA) = A / Q** (s/m) = about 30-80 for fly ash (A in $m^2$ per $m^3/s$); Gas velocity 1 - 2 m/s; plate spacing 20 - 40 cm; treatment time 2-10 s
- Efficiency is increased by increasing A (more fields), reducing gas velocity (Q), and improving w (voltage, conditioning)
- **Modified (Matts-Ohnfeldt)** equation: eta = 1 - exp[ - $(w A/Q)^k$ ], k = 0.4 - 0.6, accounts for non-ideal conditions: gas sneakage, re-entrainment

**Example:** Q = 50 $m^3/s$, w = 0.1 m/s, eta = 99% -> A = -(50 / 0.1) x ln(0.01) = 500 x 4.6 = **2300 $m^2$**.

### Dust resistivity (very important)

| Resistivity (ohm-cm) | Effect | Remedy |
|---|---|---|
| $< 10^4$ | Dust loses charge instantly; re-entrained (e.g. carbon) | Add ammonia, use baffles / wet ESP |
| $10^8 - 10^{10}$ | **Ideal** range for good collection | - |
| $> 10^{11}$ | **Back corona**: insulating layer, sparking, low efficiency (low-S coal fly ash) | Condition with $SO_3$, $H_2O$, $NH_3$, Na; hot-side ESP; pulse energisation |

- **Types:** plate-wire (most common), tubular, **wet** ESP, single-stage (charging + collection together) and **two-stage** (air conditioning), hot-side and cold-side
- **Advantages:** high efficiency (99.9%), low pressure drop (< 0.25 kPa), handles large flows and high temperature (up to 400 deg C), low operating cost, dry recovery of valuable dust
- **Disadvantages:** high capital cost, large space, sensitivity to resistivity and flow variations, produces ozone, not for explosive gases, needs skilled maintenance

**Applications: thermal power plants (fly ash), cement kilns, steel (sinter plants, blast furnaces), paper (recovery boilers), non-ferrous smelters.**

## 10. Quick Revision / Important Questions

1. Explain particle size distribution; define MMD and geometric standard deviation.
2. Explain the collection mechanisms used in particulate control.

3. Derive the efficiency and minimum particle size of a gravity settling chamber.
4. With a neat sketch explain the working and design of a cyclone; write the Lapple formulae.
5. Describe a venturi scrubber. Discuss merits and demerits of wet scrubbers.
6. Explain a baghouse; the cleaning methods and air-to-cloth ratio.
7. Explain the working of an ESP. Derive the Deutsch-Anderson equation. Effect of dust resistivity?